Practical Data Analysis Tips Pt. 2

ECON 490

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Two Components for Your Capstone Data Analysis

As part of your capstone paper, you'll include the following output:

Summary Output: *What does your data look like?*

- Summary stats table for outcome + explanatory variable(s)
- At least 2 graphs or maps for your outcome + explanatory variable(s)

Results Output: *Results in (at least) two stages*

- 1st stage is a "first pass" answer to your research question
- 2nd stage builds on and develops 1st stage along some dimension
- For most projects, this entails 2+ regression tables (some projects may differ)

Common Question: *"Is Just Doing X Good Enough?"*

Goal for Project: Be able to say/show something interesting

- This means I can't have a cookie cutter answer to what's good enough

General Rule of Thumb: Every project should entail roughly the same amount of work

- If you downloaded a single data set from a website (IPUMS, etc.), your analysis output should be relatively clean and thorough
- If you collected a lot of data manually/by hand, I take that into account

Rough Guideline: You should have more than 1 piece of output for at least 1 stage of your results (i.e., multiple regression tables/figures in one of your stages)

Common Question: *"What if I add an X variable (or swap Y)? Is that a new stage?"*

Suppose your first stage `wage ~ education` using ACS data

- Second stage is then `wage ~ education + age`
- Age is correlated with 1st stage Y/X so it's an OV—why not just include it?

In general, you should include ***all*** reasonably available explanatory variables in 1st stage, then running something substantively different in 2nd stage

- *Caveat:* If you do a lot of manual data collection, this requirement is relaxed

Miscellaneous Suggestions

Key Advice: More time spent collecting/cleaning data now means an easier project going forward—give yourself time to adjust!

You are welcome/encouraged to adjust your data & specific research question

- ***My only requirement:*** email me if you substantively change either your data or your overall research focus

If your project requires collecting/processing 2+ data sets (that aren't clean/easy to merge), your stages can be distinct parts of your analysis

- Think "two parallel stages of analysis" instead of sequential stages

Combining Data Sets

Merging data == combining data sets – why do we need to do this?

- For your projects (and most data analysis), you'll get data from ***multiple*** sources
- You combine data sets by ***merging or joining*** them together

Examples include combining:

- State-level data with individual-level survey responses
- Location-based crime data with local area demographic data
- NBA player performance data with salary and contract records

The Shape of Data Sets

When you combine two data sets, your data expands ***horizontally***

- You started with some set of columns/variables, then added new columns
- Different variables across data sets

Sometimes, you have data that you want to ***stack or append***

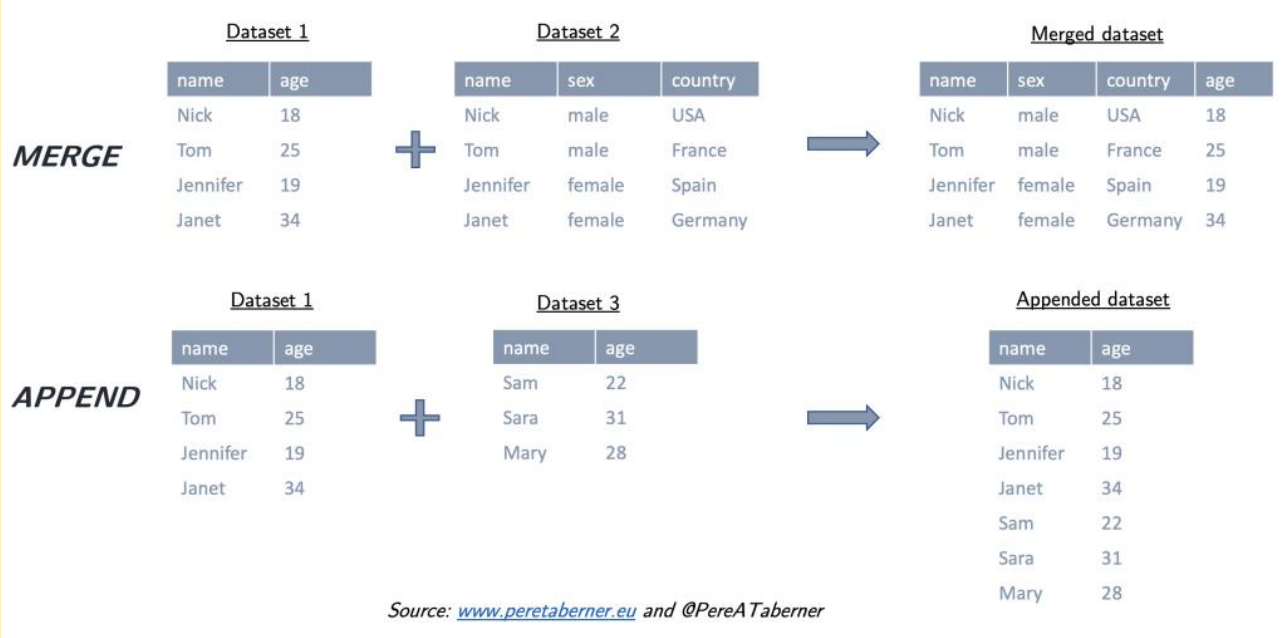
- Example: county-level pollution data for CA in 2020 and 2021
- ***Same*** variables in ***both*** data sets → append data to get panel with both years

Appending data expands your data ***vertically***

- Depending on context, merges might also, but main goal = new columns

Appending Data in R

In screenshot below, use `bind_rows()` function to ***append*** data sets 1 & 3



Before You Combine Data

Before you combine data sets, you need to understand ***data structure***

- Within both data sets, what variables uniquely identify each row?

Across both data sets, what variables will you use to link data?

- These are your “key” or “by” variables
- What is the shared structure of both data sets? Defined by ***less*** granular data

Example: Combining individual-level CPS with state-level home price data

- While CPS has individual identifiers, the ***shared structure*** is state and year, so these are the key or by variables (state-level data is less granular)

Preparing Data for Merging

Once you've identified your key variables, make sure they're ***consistent***

- Across both data sets, is each variable ***same format*** (i.e., factor, string, etc.?)
- Are spellings consistent? I.e., check "California" vs. "california" vs. "CA"

If necessary, create new, consistent versions of key variables using `mutate()`

Check you don't have duplicates across key variables

- E.g., in state-by-year ACS data, suppose you found 2 Arizona-in-2010 rows
- Drop extra row using `filter()` or combine via `group_by() + summarize()`

Join Functions in R

Given two data sets, you'll have one of the following types of merges or joins:

- ***1-to-1:*** Each observation in 1st dataset matches exactly one row in 2nd
- ***Many-to-1:*** Multiple observations in 1st dataset match to same row in 2nd
- ***Many-to-Many:*** Multiple rows match to multiple rows → avoid doing this!

Most useful join functions in R for your projects:

- `inner_join()`: Keep ***only*** matched rows in ***both*** data sets
- `left_join()`: Keep all rows from 1st data set and all matches from 2nd
- `full_join()`: Keep all rows from ***both*** data sets

After Merging Data

The most important thing to do is “sanity check” your new merged data set

- Check the number of rows in the matched data set using `nrow()`
- Does it match what you expected?

Check NAs and summary statistics for important variables using `summary()`

Example: Combine state-level crime data with state-level ACS via `inner_join()`

- This keeps matched rows, so you should have rows for 51 states (counting DC) multiplied by 5 years; if you don’t, something went wrong!

Join Examples

Inputs

DF1

ID	Value
A	123
B	769
C	475
D	978

DF2

ID	Level
A	Red
B	Blue
E	Green
F	Yellow

Join Functions

```
inner_join(DF1, DF2,
     by = "ID")
```

```
left_join(DF1, DF2,
     by = "ID")
```

Outputs

ID	Value	Level
A	123	Red
B	769	Blue

ID	Value	Level
A	123	Red
B	769	Blue
C	475	*NA*
D	978	*NA*

Quick Review of Residuals

We've defined residuals previously using the following:

$$Residual = Actual\ Y - Predicted\ Y = Y - \hat{Y}$$

$\hat{Y}$ is our "best guess" about the value of Y given our X variables

- In other words, everything in Y that our regression ***can*** explain is reflected in $\hat{Y}$
- The residual is all the "left over" variation in Y → it's what we ***can't*** explain

Important Properties of Residuals

Residuals will always have an average value of 0

- This is a general property of OLS whenever you have an intercept term
- That's why our discussion here focuses on outliers

Residuals are always conditional on a specific set of explanatory variables

- If you change your Xs, you'll get different residuals
- Residuals may be more or less "directly" interpretable given context

Residuals Example: *Using NBA Data*

Positive residuals mean actual Y is ***higher*** than we'd expect based on Xs

- Conversely, ***negative*** residuals imply actual Y is ***lower*** than we'd expect
- These deviations can provide a way of talking about your regression output

We can use data from the NBA for a concrete example

- What's the relationship between points scored (PTS) and number of shots (FGA)?
- Run the simple OLS regression `lm(PTS ~ FGA, nba.data)`

Using residuals, we can see who scores the most and least based on their shot volume

Residuals Example: *Using NBA Data*

From our model output below, each additional shot (FGA) is associated with roughly 1.3 more points (PTS)

```
Call:
lm(formula = PTS ~ FGA, data = nba.data)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.8915 -0.9854 -0.1359  0.7702  5.6119 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.47216    0.32281  -1.463    0.145    
FGA          1.34363    0.02495  53.845   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Notice part of our output is the distribution of residuals – we can explore this

Residuals Example: *Using NBA Data*

```
> # Save residuals as a new variable in our data set
> nba.data$residual.points <- resid(model.1)
> # What do our residuals look like? Let's use the summary function:
>
> summary(nba.data$residual.points)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-2.8916 -0.9854 -0.1359  0.0000  0.7702  5.6119
```

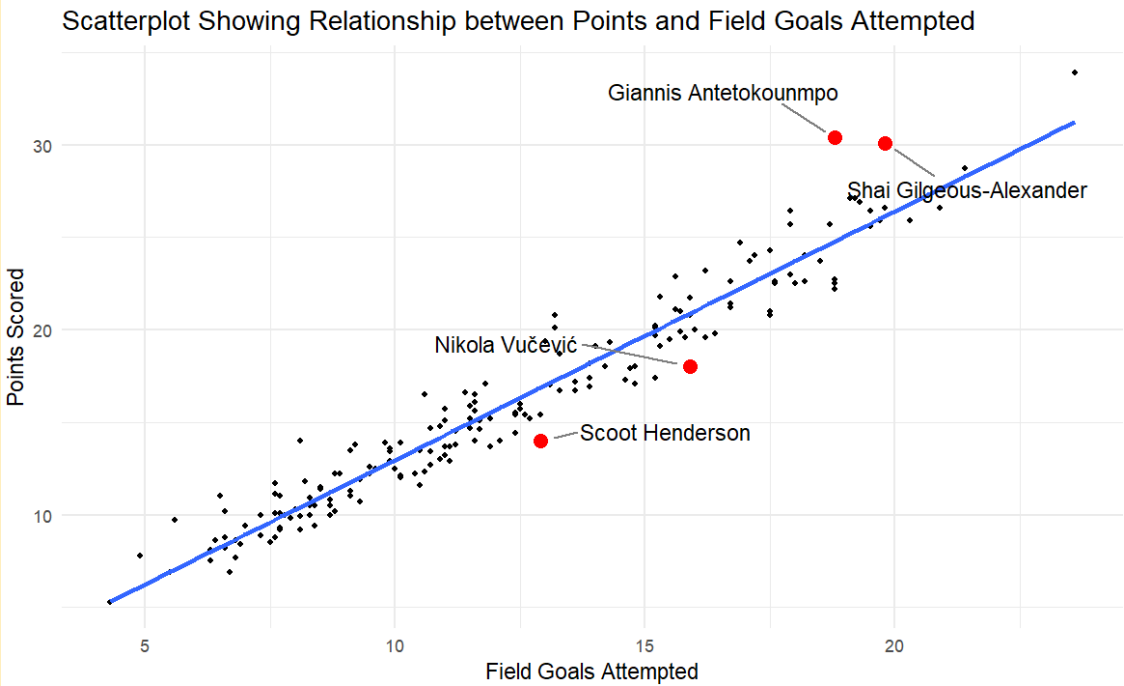
```
> # Let's see who had the largest positive values of residual points:
>
> nba.data %>%
+   arrange(desc(residual.points)) %>%
+   select(Player, Team, Pos, PTS, FGA, residual.points) %>%
+   head(10)
# A tibble: 10 × 6
   Player                 Team  Pos     PTS   FGA residual.points
   <chr>                  <chr> <chr> <dbl> <dbl>           <dbl>
 1 Giannis Antetokounmpo  MIL   PF     30.4  18.8            5.61
 2 Shai Gilgeous-Alexander OKC  PG     30.1  19.8            3.97
 3 Rudy Gobert            MIN   C      14     8.1            3.59
 4 Jimmy Butler           MIA   PF     20.8  13.2            3.54
 5 Kristaps Porziņģis     BOS   C      20.1  13.2            2.84
 6 Nikola Jokić           DEN   C      26.4  17.9            2.82
 7 Daniel Gafford         2TM   PF     11     6.5            2.74
 8 Jarrett Allen          CLE   C      16.5  10.6            2.73
 9 Luka Dončić            DAL   PG     33.9  23.6            2.66
10 Nick Richards          CHO   C       9.7   5.6            2.65
```

```
> # Let's see who had the most negative residual values:
>
> nba.data %>%
+   arrange(residual.points) %>%
+   select(Player, Team, Pos, PTS, FGA, residual.points) %>%
+   head(10)
# A tibble: 10 × 6
   Player          Team  Pos     PTS   FGA residual.points
   <chr>           <chr> <chr> <dbl> <dbl>           <dbl>
 1 Nikola Vučević  CHI   C      18    15.9           -2.89
 2 Scoot Henderson POR   PG     14    12.9           -2.86
 3 Kyle Kuzma      WAS   PF     22.2  18.8           -2.59
 4 Jordan Poole    WAS   SG     17.4  15.2           -2.55
 5 Jordan Clarkson UTA   SG     17.1  14.8           -2.31
 6 Dejounte Murray ATL   SG     22.5  18.8           -2.29
 7 Tyler Herro     MIA   SG     20.8  17.5           -2.24
 8 Cade Cunningham DET   PG     22.7  18.8           -2.09
 9 Miles Bridges   CHO   SF     21    17.5           -2.04
10 Jeremy Sochan   SAS   PF     11.6  10.5           -2.04
```

Residuals Example: *Using NBA Data*

Graph shows PTS ~ FGA relationship with outliers

Residuals are vertical difference between fitted line and ind. points



General Suggestions for Using Residuals

Opportunity to apply “qualitative” knowledge

- What characteristics do residual outliers share?
- We might not be able to measure/include this in a regression, but it gives you the chance to talk about patterns in intuitive terms

General approach for exploring residuals:

- Run regression then store residuals as a new variable
- Sort data set by residuals and explore biggest and smallest values
- Apply background knowledge to identify patterns – who stands out?

You can describe the results from this process using a table, scatterplot, or verbally